

Calculus A2 (inter)

2025 Spring

Final Exam

2025.6.13

Exam Duration: 3 Hours

Name: _____

Student ID: _____

This exam includes 7 pages (including this page) and 6 problems. Please check to see if there is any missing page, and then write down your name and student ID number on this page and the first page of your answer sheets. Also write down the initials of your name on the top of every page of your answer sheets, in case they are scattered.

This exam is open book. You are allowed to consult your textbook and notes, but NO calculators or any electronic devices. Plagiarism of all kinds are strictly forbidden and will be severely punished.

Please write down your answers to the problems in the **SEPARATE ANSWER SHEETS**, and follow the following rules:

- **Always explain your answer.** You should always explain your answers. Any problem answered with nothing but a single answer would receive no credit. At least write one sentence of explanation.
- **Write cleanly and legible.** Make sure that your writings can be read. The graders are NOT responsible to decipher illegible writings.
- **Partial credits will be given.**
- Blank spaces are provided in the exams. Feel free to use them as scratch papers. However, your formal answer has to be written in the **SEPARATE ANSWER SHEETS**, as required by the University.
- The total score of the exam is 50. If your total score exceeds 50 (there are 53 points in total), it will be recorded as 50.

Problem	Points	Score
1	8	
2	8	
3	10	
4	9	
5	9	
6	9	
Total:	53	

1. Suppose f is a twice-differentiable function on $[0, 1]$ satisfying $f(0) = 0$ and $f'(0) = 0$. Let $g = f'$ and $h = f''$. Then we first note

$$f(x) = \int_0^x g(t) \, dt,$$

$$g(t) = \int_0^t h(s) \, ds.$$

Putting these together gives

$$f(x) = \int_0^x \int_0^t h(s) \, ds \, dt.$$

So, the inverse of a second derivative is some kind of a double integral??

- (a) (2 points) In the st -plane, draw the region of integration for this double integral.
- (b) (2 points) When computing $f(x)$ via this double integral, integrate in the order dt then ds . Write the integral in this order, carefully specifying the upper bounds and lower bounds for t and s .
- (c) (2 points) In the integral from the previous part, carry out the dt integration first and simplify, to prove the following identity.

$$f(x) = x f'(x) - \int_0^x s f''(s) \, ds.$$

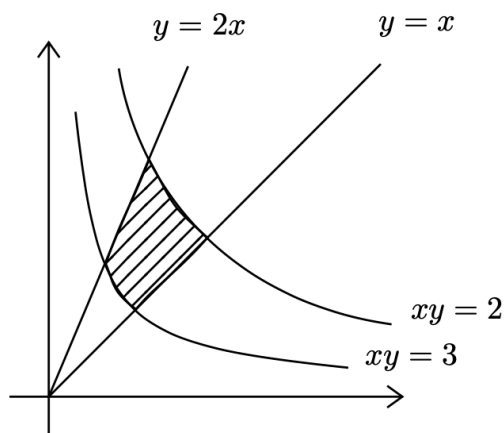
(Optional remark: this matches what you get by applying integration by parts to $f(x) = \int_0^x g(s) \, ds$.)

- (d) (2 points) Suppose $h(x) = \frac{1}{x}$ and that $f(1) = g(1) = 0$. Write the double integral for $f(x)$, use Fubini to swap the order, and evaluate it. Note the domain is now $(0, \infty)$ rather than $[0, 1]$.

2. We have the planar vector field

$$\mathbf{F} = \begin{bmatrix} -2y^3 \\ 6xy^2 \end{bmatrix},$$

and γ is the boundary of the shaded region below, oriented counterclockwise.

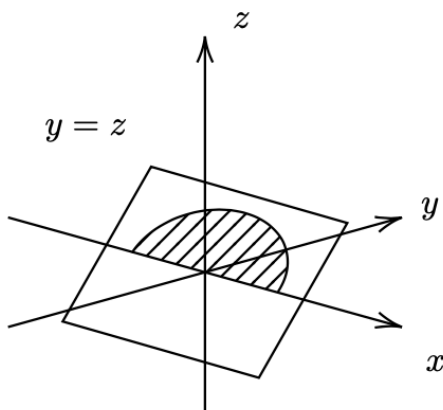


- (a) (2 points) Let $u = xy$ and $v = \frac{y}{x}$. Find the relation between $dA_{x,y}$ and $dA_{u,v}$ in terms of u, v .
- (b) (2 points) Compute the work integral $\int_{\gamma} \mathbf{F} \cdot d\gamma$, with γ counterclockwise.
- (c) (2 points) Compute the flux $\int_{\gamma} \mathbf{F} \cdot \mathbf{n} \, ds$, taking outward as positive.
- (d) (2 points) Decompose $\mathbf{F}$ into two fields, one with zero curl everywhere and one with zero divergence everywhere. (Hint: find ϕ so that $\nabla\phi$ has the same divergence as $\mathbf{F}$. Then $\nabla\phi$ is curl-free and $\mathbf{F} - \nabla\phi$ is divergence-free.)

3. In $\mathbb{R}^3$, let γ be the circle given by intersecting the sphere $x^2 + y^2 + z^2 = 1$ with the plane $y = z$, for $z \geq 0$. Its orientation goes from $(-1, 0, 0)$ to $(1, 0, 0)$. Let

$$\mathbf{F} = \begin{bmatrix} z + x^2 + 2xy + yz \\ x^2 - x + e^y + xz \\ xy \end{bmatrix}.$$

We want to compute the work $\int_{\gamma} \mathbf{F} \cdot d\gamma$, but γ is hard to parametrize and $\mathbf{F}$ is quite messy.



- (a) (2 points) Compute $\nabla \times \mathbf{F}$.
- (b) (2 points) Let S be the disk cut out by the intersection of the solid sphere $x^2 + y^2 + z^2 \leq 1$ with the plane $y = z$, for $z \geq 0$. Let η be the straight line segment from $(1, 0, 0)$ to $(-1, 0, 0)$. If $\partial S = \gamma + \eta$, find the unit normal $\mathbf{n}$ on S (consistent with the boundary orientation).
- (c) (2 points) For the oriented surface S , compute the flux of $\nabla \times \mathbf{F}$ through S .
- (d) (2 points) Compute $\int_{\gamma} \mathbf{F} \cdot d\gamma$.
- (e) (2 points) Compute

$$\int_{\gamma} e^y \cos z \, dx + x e^y \cos z \, dy - x e^y \sin z \, dz.$$

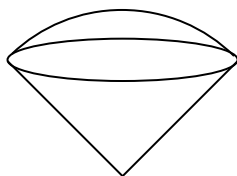
4. Let

$$\mathbf{F} = e^{x^2+y^2+z^2} \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

and let U be the region between the cone $z^2 = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 1$ for $z \geq 0$. We wish to compute

$$\int_U \nabla \cdot \mathbf{F} \, dV$$

over this “ice-cream” region.



(a) (2 points) Compute $\nabla \cdot \mathbf{F}$.

(b) (3 points) Switch to spherical coordinates to write out the integration formula

$$\int_U \nabla \cdot \mathbf{F} \, dV$$

and carry out the ϕ, θ integrals (you may leave the ρ -integral uncomputed).

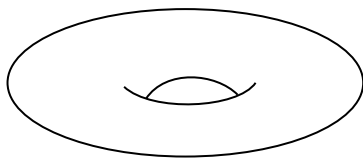
(c) (2 points) Compute $\int_U \nabla \cdot \mathbf{F} \, dV$ using the divergence theorem.

(d) (2 points) Let C be the conical part of ∂U (the part below U , and the normal direction should be given from ∂U). Compute

$$\int_C z^2 \, dy \wedge dx.$$

5. Consider the torus parametrized by

$$\Phi(u, v) = \begin{bmatrix} (2 + \cos v) \cos u \\ (2 + \cos v) \sin u \\ \sin v \end{bmatrix}.$$



(a) (2 points) Compute the normal direction $D_u \Phi \times D_v \Phi$ and decide whether it points outward or inward, with a brief justification.

(b) (3 points) Let S^+ be the part of the torus with $z \geq 0$. Assuming uniform density, find the centroid of S^+ .

(c) (2 points) Suppose

$$\mathbf{F} = \begin{bmatrix} -y + xz \\ x + yz \\ e^{xyz} \end{bmatrix}.$$

Compute the flux of $\nabla \times \mathbf{F}$ through S with the outward normal direction, and through S^+ with the upward normal direction.

(d) (2 points) Let V be the solid torus enclosed by S . Compute

$$\int_V |z| \, dV.$$

6. Consider the series (with a_n its n th term):

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots,$$

and let s_n be its n th partial sum.

Next, consider the series (with b_n its n th term):

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \dots,$$

and let S_n be its n th partial sum.

- (a) (3 points) Are these two series conditionally convergent? Absolutely convergent? Explain which test you use. (Hint: for S_n , look at the termwise product of

$$(1, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{7}, \dots) \quad \text{and} \quad (1, 1, -2, 1, 1, -2, \dots).$$

If you forget the test's name, describe which properties guarantee convergence.)

- (b) (2 points) Let $s = \lim s_n$ and $S = \lim S_n$. Prove $S \neq s$. (The 1954 Putnam asked to compute S . We don't require that.)

- (c) (2 points) With $b_0 = 0$, find the radius of convergence and describe the interval of convergence of

$$\sum b_n(x-1)^n.$$

Are either endpoint included?

- (d) (2 points) Let f be 2π -periodic with $f(x) = \pi - x$ on $[0, 2\pi)$. Its Fourier series is

$$S(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + \sum_{n=1}^{\infty} b_n \sin(nx).$$

Find formulas for all a_n, b_n and compute $S(0)$.